26900161 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECFS300 - REMOTE SENSING NETWORK

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Explain the basic principle of Planck's Law. [U] / [CO1]
2. Explain how different regions of the electromagnetic spectrum provide information about different surface features. [U] / [CO1]
3. State any two limitations of Earth resource satellite. [K] / [CO2]
4. Define an airborne observation platform. [K] / [CO2]
5. Differentiate between active and passive sensors. [U] / [CO2]
6. List any two examples of ground observation platform. [K] / [CO2]
7. Define Thermal Sensor. [K] / [CO2]
8. Identify any two examples of transformed data used in remote sensing. [U] / [CO3]
9. Explain how visual information can be understood from an image. [U] / [CO3]
10. List the different steps involved in Digital Image Processing. [K] / [CO3]

11. Identify the purpose of shoreline-change detection using remote sensing. [U] / [CO4]

12. Define forest-cover mapping and state one application. [K] / [CO4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

13. Explain the nature of electromagnetic radiation and discuss its importance in Remote Sensing. [U] / [CO1]

14. Describe a platform. Explain the different types. [U] / [CO2]

15. Explain the applications of remote sensing in land-use and land-cover mapping. [U] / [CO4]

16. Illustrate how multi-spectral, multi-temporal, and multi-sensor data could be combined to study an environmental change problem. [A] / [CO3]

17. Explain ground truth importance in interpretation and accuracy assessment of remote sensing data. [U] / [CO3]

18. Illustrate the importance of data integration, analysis, and presentation in the decision-making process. [A] / [CO3]

19. Present remote sensing techniques for geomorphological studies and landform mapping. [A] / [CO4]

20. Discuss about the advantages of using remote sensing for urban studies. [U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

21. Describe the major milestones in the history of Remote Sensing and explain how the development of different technologies contributed to modern Remote Sensing. [U] / [CO1]

22. Explain the limitations of remote sensing data analysis in detail. [U] / [CO3]

23. Discuss the applications of remote sensing in agriculture. [U] / [CO4]

[10x2 = 20]